MAHFUZUR RAHMAN

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LinkedIn Profile • Portfolio

EDUCATION

Bachelor of Science in Aeronautical Engineering April 2022 – May 2026

Military Institute of Science and Technology (MIST) Dhaka, Bangladesh

CGPA: **3.80 / 4.00** | Dean's List of Honour: Spring 2023 & Spring 2024

Relevant Coursework: Applied Aerodynamics, Computational Fluid Dynamics, Aircraft Structures, Heat Transfer, Advanced Materials Processing, Aerospace Propulsion, High Speed Aerodynamics, Numerical Analysis, Control Systems.

UNDERGRADUATE THESIS PROJECT

Thesis Title:	**Thermomechanical Behavior of Fiber-Reinforced Composite Materials**
Supervisor:	Dr. Shaikh Reaz Ahmed, Professor, Department of Mechanical Engineering, Bangladesh University of Engineering and Technology (BUET).
Synopsis:	• Developed a displacement-function-based finite difference numerical solver in MATLAB for coupled thermomechanical analysis of orthotropic composite structures (Al-B_4C MMC). • The solver resolves the complete thermoelastic field i.e. stress, displacement, and temperature distribution with accuracy exceeding ABAQUS FEM benchmarks under equivalent boundary conditions.

RESEARCH INTERESTS

1. Computational Fluid Dynamics (CFD) and Numerical Modelling of aerospace flow fields.
2. Thermomechanical Analysis of Composite and Metallic Structures.
3. Structural Analysis and Design of Aerospace Structures.
4. Fixed-Wing UAV Aerodynamic Design and Manufacturing.

COMPETITION PROJECTS

AIAA 30th Annual Design/Build/Fly (DBF) 2026 – Team Lead Wichita, Kansas, USA

UAV: HeisenBird | Theme: Banner Towing Bush Plane

- Directed the full design/build/fly lifecycle of a 42-member team; led aerodynamic sizing in XFLR5 and AVL, structural design in SolidWorks
- Ranked 58th worldwide out of 175 competing universities; achieved 17th worldwide in Design Report (1st in Asia) and 18th worldwide in Proposal Report.

AIAA 29th Annual Design/Build/Fly (DBF) 2025 – Manufacturing Lead Tucson, Arizona, USA

UAV: Akashtori 2.0 | Theme: X-1 Supersonic Flight Program

- Managed composite manufacturing workflow including glass-fibre vacuum-bag layup, rib fabrication, and structural assembly; ensured all airframe components met dimensional and weight budget requirements.
- Supported aerodynamic analysis in XFLR5 and AVL; performed structural assessments in ANSYS and ABAQUS for wing spar and fuselage frame under critical load cases.
- Ranked 39th worldwide out of 112 competing universities; secured 1st in Asia for Total Flight Mission Score.

INTERNSHIP EXPERIENCE

Industrial Trainee (May – June 2025) Organization: **US-Bangla Airlines Ltd.**

CAMO & AMO departments, ATR 72-600 and Boeing 737-800 fleets. Reviewed Airworthiness Directives (ADs), MEL, MCC, and PPC; observed NDT procedures and SRM documentation; gained exposure to SMS processes within a Part-145 approved environment.

LEADERSHIP AND PROJECT MANAGEMENT

Vice President – Administration (Jun 2025 – Jun 2026)	Organization**: MIST Aeronautics and Astronautics Club (MAAC)** Supervised administrative operations; coordinated inter-departmental correspondence and the annual national aeromodelling competition.

Department Representative (Jul 2025 – Jun 2026)	Organization**: Department of Aeronautical Engineering, MIST** Liaison between student body and faculty for academic scheduling, curriculum feedback, and co-curricular event coordination.
Executive – Administration (Sep 2024 – Jun 2025)	Organization**: MIST Aeronautics and Astronautics Club (MAAC)** Organized club workshops on aircraft modelling and flight simulation
Director of Event Management (Jun 2024 – Jun 2025)	Organization**: MIST Drama and Film Society (MDFS)** Oversaw planning and execution of theatrical and film-screening events; managed a 20-member event crew.
Event Manager (May 2023 – Sep 2024)	Organization**: MIST Aeronautics and Astronautics Club (MAAC)** Planned and executed aviation-themed events and technical seminars; managed venue logistics and volunteer coordination.

ONLINE COURSES AND CERTIFICATIONS

MATLAB Onramp	Organization: **MathWorks**
Simulink Onramp	Organization: **MathWorks**
Python	Organization: **Kaggle**
EF SET English Certificate – C1 Advanced (CEFR)	Organization: **EF SET**
Prompt Engineering for Everyone	Organization: **Cognitive Class**

AWARDS AND ACHIEVEMENTS

2026	**17th out of 175** teams in Design Report in **Design/Build/Fly (DBF)** Competition	AIAA
2025	**39th out of 112** teams in **Design/Build/Fly (DBF)** Competition	AIAA
2025	**MIST Merit Scholarship** - Spring 2025, for ranking **3rd** in the class based on academic performance.	MIST
2023 - 2024	**MIST Dean's List of Honour**, for earning a GPA greater than 3.75 in each academic year	MIST
2021	**Board Scholarship** (General) in HSC	Chattogram Education Board
2019	**Board Scholarship** (Talentpool) in SSC	Chattogram Education Board

SKILLS

Programming Language:	**Python, MATLAB, C**
Design and Simulation:	**SolidWorks, XFLR5, AVL, ANSYS, ABAQUS**
Numerical Methods:	**Finite Difference Method (FDM), Finite Element Method (FEM), Numerical Analysis**
Manufacturing:	**Composite Fabrication (vacuum-bag layup, glass-fibre laminate), Structural Assembly**
Aviation MRO:	**CAMO/AMO, MEL, Airworthiness Directives, NDT, SRM, SMS**
Documentation:	**LaTeX, Microsoft Office Suite**

LANGUAGES

Bangla: **Native**

English: **Fluent**

REFERENCES

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